

Project Based Learning

On

RCC Framed Structure Detailing Using AutoCAD.

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RCC FRAMED STRUCTURE DETAILING USING AUTOCAD

PROJECT REPORT

ABSTRACT

Reinforced Cement Concrete (RCC) framed structures are among the most widely used structural systems in modern buildings because of their strength, durability, versatility, and economical construction. AutoCAD is a widely adopted computer-aided drafting application that enables civil and structural engineers to prepare accurate, clear, and well-organized drawings. This project focuses on the preparation of RCC framed structure detailing using AutoCAD for a G+1 residential building. The drawing set covers foundation layout, column reinforcement, beam reinforcement, slab reinforcement, sectional views, dimensions, annotations, and construction-related notes. Appropriate drawing layers, object snaps, modification commands, dimensioning tools, and plotting settings are used to improve drafting quality. The project demonstrates how CAD-based detailing can reduce manual drafting errors, simplify revisions, improve communication among project participants, and produce professional construction documentation. The prepared drawings are intended for academic demonstration and must be checked by a qualified structural engineer before actual construction.

PROJECT CONTENTS

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2. Foundation Layout Plan
3. RCC Beam and Slab Reinforcement
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9. Advantages
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INTRODUCTION

Reinforced Cement Concrete (RCC) is a composite construction material in which concrete and steel reinforcement act together to resist structural forces. Concrete is particularly effective in resisting compression, while steel reinforcement provides tensile resistance and contributes to ductility and crack control. RCC framed construction is commonly used for residential, commercial, institutional, industrial, and other building applications.

In a framed structure, loads from slabs are transferred to beams, from beams to columns, and from columns to foundations and finally to the supporting soil. The structural drawing package communicates the arrangement and detailing of these components to the construction team.

AutoCAD provides a precise environment for producing structural layouts, reinforcement drawings, sections, dimensions, notes, schedules, and plotting sheets. It also allows drawings to be modified quickly when design changes are required.

This project illustrates the preparation of RCC framed structural drawings for a G+1 residential building with a plan size of 12 m × 9 m. The report emphasizes drafting methodology and detailing rather than a complete structural design calculation.

PROJECT OBJECTIVES

- Prepare clear RCC foundation, beam, slab, and column detailing drawings.
- Apply accurate dimensions, reinforcement notation, labels, and section references.
- Use AutoCAD layers, object snaps, drawing and modification commands effectively.
- Develop a professional drawing set suitable for academic structural-drafting work.
- Understand the relationship between structural components and construction documentation.

DRAWING – 1: FOUNDATION LAYOUT PLAN

The foundation layout establishes the location and identification of columns and isolated footings with respect to the building grid. The proposed building footprint is 12,000 mm long and 9,000 mm wide. Column centre lines are arranged on a regular structural grid to support the superstructure.

DESIGN / DRAFTING DATA

Item	Dimension / Specification
Building Size	12,000 × 9,000 mm
Isolated Footing	1,800 × 1,800 mm
Footing Thickness	450 mm
PCC Thickness	100 mm
Concrete Grade	M25
Steel Grade	Fe500
Bottom Reinforcement	12 mm @ 150 mm c/c both ways
Column Size	300 × 450 mm

AUTOCAD DRAWING INCLUDES

- Grid lines
- Footing layout
- Column positions

- Centre lines
- Dimensions
- Footing numbering
- Section marks

Typical drafting sequence: establish the building grid, locate column centre points, draw footing outlines using the specified dimensions, mark column positions, add dimensions and identifiers, and finally place section references and notes.

DRAWING – 2: RCC BEAM

Beam and slab detailing drawings communicate member sizes and reinforcement arrangement. The drawing should clearly distinguish top reinforcement, bottom reinforcement, stirrups, slab main reinforcement, distribution reinforcement, concrete cover, bar marks, and sectional references.

BEAM DETAILS

Item	Size / Specification
Beam Size	230 × 450 mm
Main Bars	4 Nos. 16 mm
Top Bars	2 Nos. 16 mm
Bottom Bars	2 Nos. 20 mm
Stirrups	8 mm @ 150 mm c/c
Concrete Cover	25 mm

DRAWING – 3: COLUMN REINFORCEMENT DETAIL

Columns transfer axial and bending effects from beams and slabs toward the foundations. The column reinforcement drawing shows the member size, longitudinal reinforcement, lateral ties, clear cover, lap information, and cross-sectional arrangement.

COLUMN REINFORCEMENT DATA

Item	Value
Column Size	300 × 450 mm
Longitudinal Bars	8 Nos. 20 mm
Lateral Ties	8 mm @ 150 mm c/c
Lap Length	1000 mm
Clear Cover	40 mm
Concrete Grade	M25

DRAWING CONTENT

- Front elevation
- Cross-section
- Reinforcement arrangement
- Tie spacing
- Dimensions

Important: reinforcement sizes, lap lengths, spacing, covers, anchorage, and detailing requirements shown in an academic drawing must be verified against the applicable design calculations and current codes before construction.

DRAWING DETAILING WITH MEASUREMENTS

The following dimensional data are used as the basis for preparing the architectural/structural drafting layout and coordinating the RCC framing drawings.

BUILDING DIMENSIONS

Description	Dimension
Overall Length	12,000 mm
Overall Width	9,000 mm
Living Room	4,500 × 4,000 mm
Bedroom	3,600 × 3,600 mm
Kitchen	3,000 × 3,000 mm
Toilet	1,800 × 2,100 mm
Staircase Width	1,500 mm
External Wall	230 mm
Internal Wall	115 mm
Plinth Height	600 mm
Floor Height	3,000 mm
Roof Slab Thickness	150 mm

DIMENSIONING PRACTICE

- Use millimetres for structural dimensions unless otherwise stated.
- Maintain consistent dimension text, arrowheads, extension lines, and units.
- Dimension overall building sizes before internal room or member dimensions.
- Use centre lines for columns and structural grids to avoid ambiguity.
- Keep reinforcement notes close to the relevant structural member or section.

AUTOCAD DRAWING TOOLS USED

AutoCAD commands are grouped according to their drafting purpose. Using a consistent command workflow improves accuracy and makes structural drawings easier to revise.

DRAWING TOOLS

- LINE
- POLYLINE
- CIRCLE
- ARC
- RECTANGLE
- POLYGON
- HATCH
- ELLIPSE
- SPLINE• DONUT

MODIFY TOOLS

- MOVE
- COPY
- ROTATE
- MIRROR
- OFFSET
- TRIM
- EXTEND
- SCALE
- STRETCH
- ARRAY
- FILLET
- CHAMFER
- BREAK
- JOIN
- EXPLODE

ANNOTATION TOOLS

- DIMENSION
- MULTILINE TEXT
- LEADER
- TABLE
- REVISION CLOUD

LAYER MANAGEMENT

- Layer Properties
- Layer Freeze
- Layer Lock
- Layer Color
- Lineweight

OBJECT SNAP (OSNAP)

- Endpoint
- Midpoint
- Centre
- Intersection
- Perpendicular
- Tangent

PLOTTING

- Layout
- Viewport
- Plot Style
- PDF Export
- Scale Setting

AUTOCAD DRAFTING METHODOLOGY

A systematic drafting workflow is followed to obtain a clean and coordinated structural drawing set.

- Set drawing units and establish the required drawing scale.
- Create separate layers for grids, columns, beams, slabs, reinforcement, dimensions, text, hatching, andconstruction notes.
- Draw the structural grid and establish accurate centre lines.
- Place columns at grid intersections and draw the foundation layout.
- Draft beam and slab framing plans using the specified member dimensions.
- Prepare enlarged reinforcement sections for beams, slabs, columns, and footings.
- Add bar marks, reinforcement notes, dimensions, section symbols, and general notes.
- Check geometry, alignment, dimensions, layer visibility, and lineweights.
- Create paper-space layouts, set viewports and scales, and prepare the final plotted sheet.
- Export the completed drawings to PDF for documentation and submission.

QUALITY-CONTROL CHECKLIST

- All structural members have unique identification marks.• Dimensions do not overlap important drawing geometry.
- Reinforcement notes agree with the intended detailing schedule.
- Layers and lineweights are consistent throughout the drawing set.
- Section references correspond to the correct locations.
- Plot preview is checked before final PDF export.

TECHNOLOGY USED

SOFTWARE

- AutoCAD 2024

OPERATING SYSTEM

- Windows 10 / Windows 11

DRAFTING / DESIGN REFERENCES

- IS 456:2000 – Plain and Reinforced Concrete – Code of Practice
- IS 875 – Design Loads for Buildings and Structures
- SP 34 – Handbook on Concrete Reinforcement and Detailing
- IS 13920 – Ductile Design and Detailing of Reinforced Concrete Structures Subjected to Seismic Forces

HARDWARE

- Intel Core i5 Processor
- 8 GB RAM
- 512 GB SSD
- Dedicated Graphics Card

Note: The listed hardware represents a suitable academic drafting configuration. Actual requirements vary with drawing complexity and software version.

APPLICATIONS

- Residential Buildings
- Commercial Buildings
- Hospitals
- Schools
- Industrial Buildings
- Shopping Complexes
- Apartment Buildings
- Office Buildings
- Structural Consultancy• Construction Projects

ADVANTAGES

- High drafting accuracy
- Faster drawing preparation
- Easy modification of drawings
- Better reinforcement detailing
- Reduced human errors
- Professional documentation
- Improved construction quality
- Better communication among engineers
- Time-saving
- Cost-effective drafting

ROLE OF CAD IN STRUCTURAL DOCUMENTATION

CAD-based structural drafting supports coordination between design information and construction documentation. A well-organized drawing set makes it easier to identify member sizes, reinforcement arrangements, dimensions, and construction notes. Digital drawings can also be revised and reproduced efficiently, which is especially useful when changes occur during design development.

CONCLUSION

The project demonstrates the preparation of RCC framed structural drawings using AutoCAD. The software provides an accurate and organized platform for creating foundation layouts, beam and slab reinforcement drawings, column reinforcement details, sections, dimensions, annotations, and construction notes. Proper use of layers, object snaps, drawing commands, modification commands, and plotting settings improves the quality and consistency of the final drawing set.

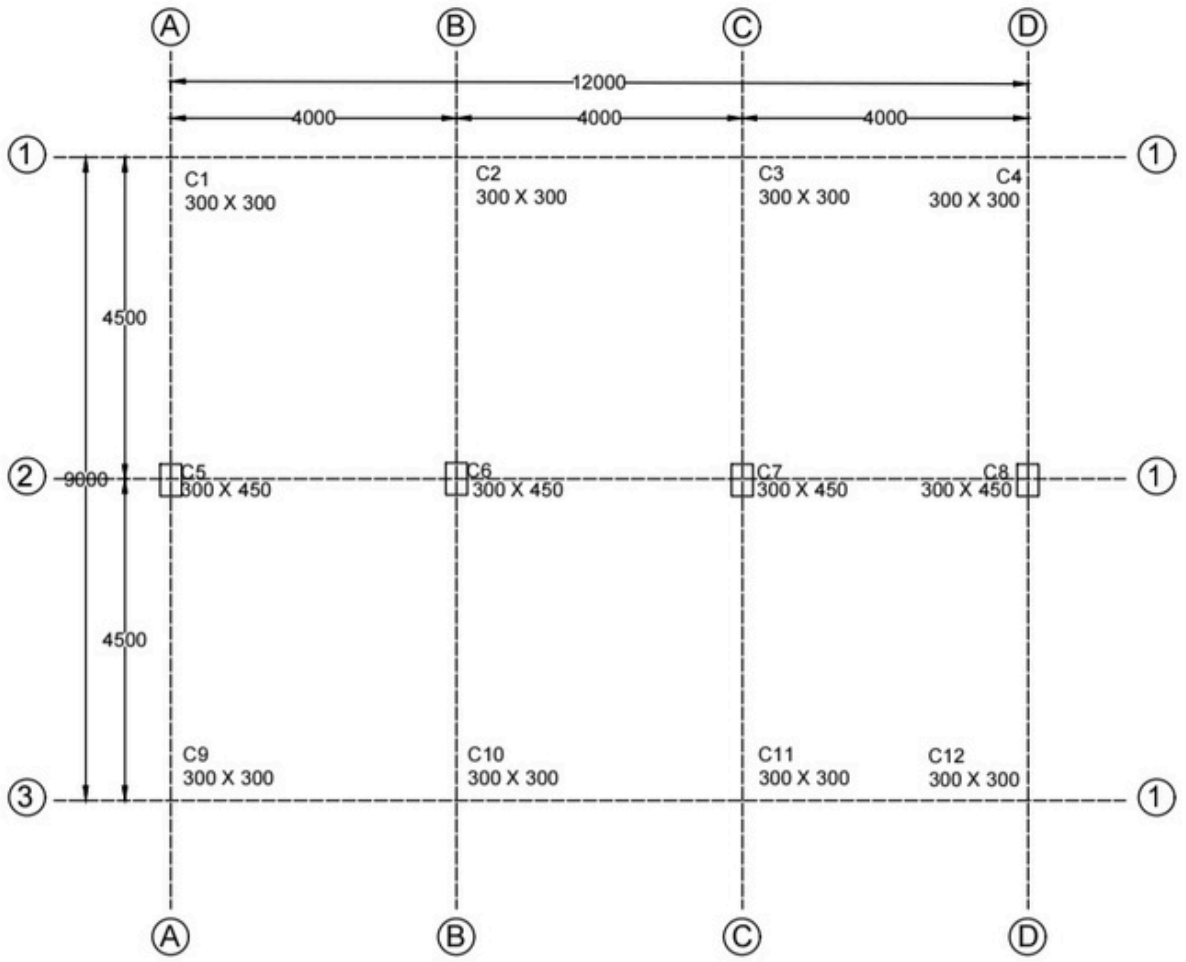
The use of AutoCAD can reduce manual drafting errors, simplify revisions, save time, and improve communication among engineers, draftsmen, contractors, and other construction professionals. The project also provides practical exposure to structural drafting conventions and reinforcement representation.

The dimensions and reinforcement information in this academic report are drafting/design data supplied for the project and should not be treated as a construction-ready structural design. Final structural design, reinforcement adequacy, soil suitability, seismic requirements, and construction drawings must be verified and approved by a qualified structural engineer in accordance with the applicable and current standards.

REFERENCES

- Bureau of Indian Standards, IS 456:2000, Plain and Reinforced Concrete – Code of Practice.

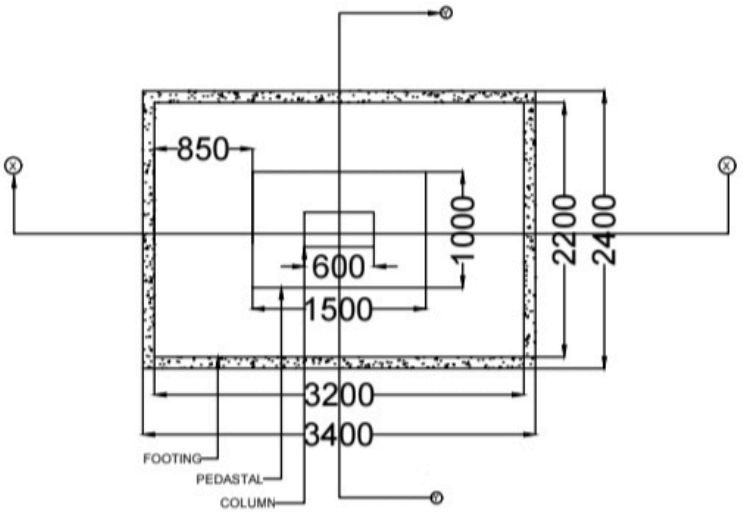
- Bureau of Indian Standards, IS 875, Design Loads (Other than Earthquake) for Buildings and Structures.
- Bureau of Indian Standards, SP 34, Handbook on Concrete Reinforcement and Detailing.
- Bureau of Indian Standards, IS 13920, Ductile Design and Detailing of Reinforced Concrete StructuresSubjected to Seismic Forces.
- Autodesk, AutoCAD documentation and user guidance.



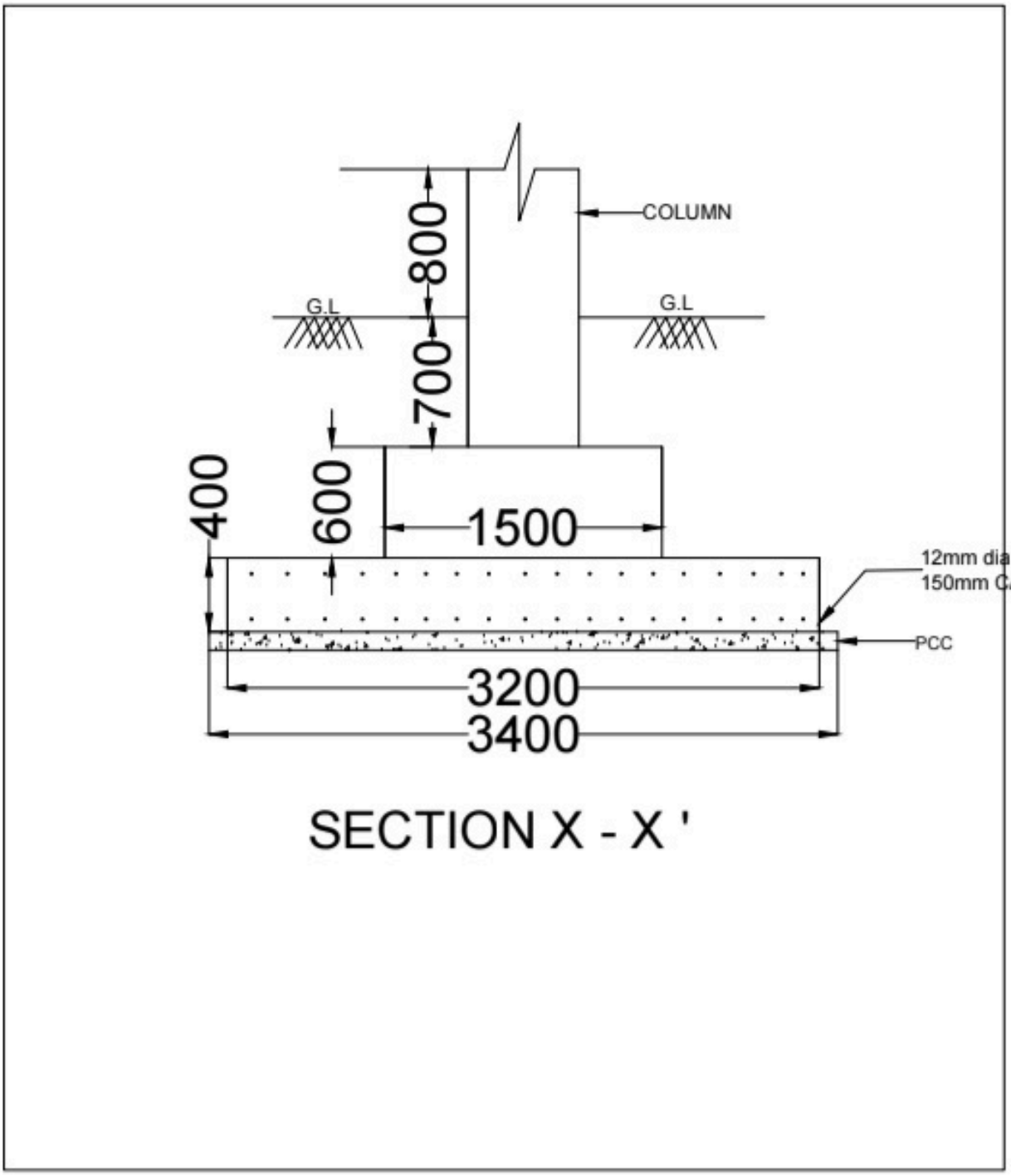
RCC COULMN LAYOUT PLAN

(All dimensions are in mm)

ISOLATED FOOTING TOP VIEW



PLAN


COLUMN
800
G.L
G.L
700
600
400
1500
12mm dia of Bars
150mm C/C RCC
PCC
3200
3400

SECTION X - X '